# Changelog — Electric Cable Shovel Digital Twin (M1, M2, M3)

## [2026-11-06] — Milestone M2 Completion (P05–P08 & Formal SVD) + Milestone M3 (v2.0.0-PRELIMINARY) Integration

### Added

- **Literature Mapping**: Created research_documentation/literature_to_m2_m3_mapping.md (LITERATURE_TO_M2_M3_MAPPING.md) synthesizing all 58 BibTeX entries in papers_library.bib and the 11 estimation-theory additions in papers.md.
- **M2 Current State Audit & Register**: Created M2_CURRENT_STATE_AUDIT.md and M2_EXPERIMENT_REGISTER.xlsx documenting all 28 .mat archives (with SHA-256 hashes and git provenance) across MS02P00–MS02P08.
- **M2 Formal SVD Identifiability (P06)**: Created M2_FORMAL_IDENTIFIABILITY_REPORT.md, M2_P06_GEOMETRY_STATION_REPORT.md, and M2/scripts/m2_formal_identifiability_svd.m computing unscaled, unit-column-norm, and physically scaled SVD condition numbers \kappa_2(S), VIF, and subspace angles \theta_{\min} across 16 trajectories, and demonstrating how joint 2-channel (\tau_3, f_4) estimation reduces narrow-band conditioning from 497.71 to 3.77 (\theta_{\min} = 29.70^\circ).
- **M2 Configuration Sweep (P05)**: Created M2_P05_CONFIGURATION_SWEEP_REPORT.md and M2/scripts/m2_p05_config_sweep.m proving \Vert{}\tilde{Q}(q)\Vert{}_2 / \Vert{}F\Vert{}_2 = 1.0000 across all q_3 \in [0^\circ, 90^\circ] and verifying quadrant-binned residuals on M2_100kN_0p1111Hz_sp1p50.mat.
- **M2 Model Perturbation Fingerprint Library (P07)**: Created M2_P07_MODEL_PERTURBATION_REPORT.md and M2/scripts/m2_p07_model_perturbation.m evaluating 15 parameter perturbations (\Delta M_d, \Delta I_{zz}, \Delta c, \Delta f_v, F_c) and verifying exact least-squares basis attribution on actual Simscape trajectories.
- **M2/M3 Cross-Trajectory Held-Out Validation (P08)**: Created M2_P08_HELD_OUT_VALIDATION_REPORT.md and M2/scripts/m2_p08_held_out_validation.m validating both M2 algebraic recovery and M3 30\ \mathrm{Hz} momentum observers across all 20 correct-station Simscape .mat archives.
- **M3 Workspace Organization & Code-to-Report Audit**: Organized M3/00_README through M3/10_ARCHIVE with zero file deletions, documented in M3_FILE_MIGRATION_REGISTER.xlsx, M3_CODE_AUDIT_REPORT.md, M3_VALIDATION_PROTOCOL.md, m3_observer_suite.m (v2.0.0-PRELIMINARY), and

M3_LaneB_Momentum_Observer_Report.pdf (v2.0.0-PRELIMINARY).

# CHANGELOG — ShovelSimulator.slx

One entry per model version. The authoritative .slx is the one flagged (AUTHORITATIVE); old versions live in git history. Deviation 2026-07-15: superseded file copies are temporarily kept in the working tree under a no-delete policy; archive pass deferred.

## v5 — 2026-07-15 — sensed-acceleration wiring restored (AUTHORITATIVE: ShovelSimulatorr2025b.slx)

- AUTHORITATIVE MODEL = ShovelSimulatorr2025b.slx (name kept per no-delete policy; canonical rename to ShovelSimulator.slx deferred — MUST happen before any M2 edit begins).
- Defect found and fixed this date: To Workspace block was named q_3ddot (missing underscore) → script silently fell back to gradient() on the revolute. Renamed to q_3_ddot.
- Wiring verified: q_3_ddot + d_4_ddot SENSED, $R^2$ = 1.000000 both equations, blind ID exact (validate_and_identify v5.1, self-describing results).
- ShovelSimulator.slx in this folder = PRE-WIRING version, kept unchanged.
- Stray copies preserved per no-delete policy: scripts/ShovelSimulator_1_.slx (typo-era Downloads copy), model/FourDOFShovelSimulator.slx (+.slx.r2024a autosave; origin to be documented by owner — see research log 2026-07-15 open questions).

## v4 — 2026-07-06 — planar geometry made consistent

- Rigid Transform1 rotation: reverted to None (superseded by gravity tilt).
- Middle Rigid Transform (the 1.5 m one): rotation StandardAxis +Y 90° → prismatic slide axis made RADIAL (was axial along the spin axis).
- Dipper Handle: COM [0 0 −2.82] (moved onto slide axis); MomentsOfInertia [287900 0 0] (moved to the axis parallel to spin).
- Mechanism Configuration: gravity [0 −9.80665 0] — in-plane ("tilt gravity, not the mechanism" — standard planar-model practice).
- Net lever arm: r = 1.5 + d4 − 2.82 = d4 − 1.32.
- Post-v4 wiring (2026-07-07): revolute 'b' → q_3_ddot, prismatic 'a' → d_4_ddot (each via PS-Simulink → To Workspace). Stop time 30 s.
- Validation state: BOTH equations $R^2$ = 1.000000 with sensed accelerations.
- LIMITATION to remember: planar trick valid only while the subsystem is planar; the 4-DOF extension with swing needs genuine 3-D frames.

## v3 — 2026-07-06 — Rasuli Table II parameters adopted

- Dipper mass 90,300 → 54,300 kg; COM −5 → −2.82 m; Izz 839,000 → 287,900.
- Reason: old values traced to NO source (checked Frimpong 2004/06/08 and all project

docs — absent); Table II values are published and citable.
- Superseded the same day by v4 (geometry was still wrong).

# v2 — 2026-07-06 — friction + excitation fixes ("withFriction")

- Revolute DampingCoefficient: 0 → 66.322512 N·m/(deg/s) (= 3,800 N·m·s/rad). (First patch used 5.934 = 340×π/180 from a MISREAD of Table II; corrected.)
- Prismatic DampingCoefficient: 0 → 1,473 N/(m/s) (same correction history).
- Sine Wave1 (d4) amplitude 0.25 → 2 m — fixed regressor ill-conditioning (cond(Y) scaled ~1e4–1e5 → ~85).
- Rigid Transform1 rotation StandardAxis +X 90° added (partial axis fix — fixed gravity coupling into the revolute but left the slide axial; fully superseded by v4).

# v1 — original — as first built

- 2-DOF chain: World → Boom Base → Revolute (Rz) → Rigid Transform [1.5 0 0] → Prismatic (Pz) → Dipper Handle.
- Dipper: 90,300 kg, COM [−5 0 0], Izz 839,000 (provenance unknown).
- Gravity [0 0 −9.80665]; both joint axes on Simscape default Z → spin axis vertical (the root geometry bug), slide axial.
- Sine inputs: q3 45°±15°, d4 8.75±0.25 m, both ω = π/10; stop 10 s; solver VariableStepAuto.